6.6.5 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to help them understand how to determine if a sequence of transformations results in similar figures.

Unit 6, Lesson H1: Proving All Circles are Similar

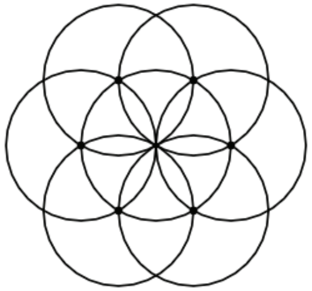
Benchmark

MA.912.GR.6.5 Apply transformations to prove that all circles are similar.

Learning Target

☐ I can justify that two circles are similar and conclude that any circle can be mapped to another circle using similarity transformations.

6.H1.1 Warm-Up: Notice and Wonder



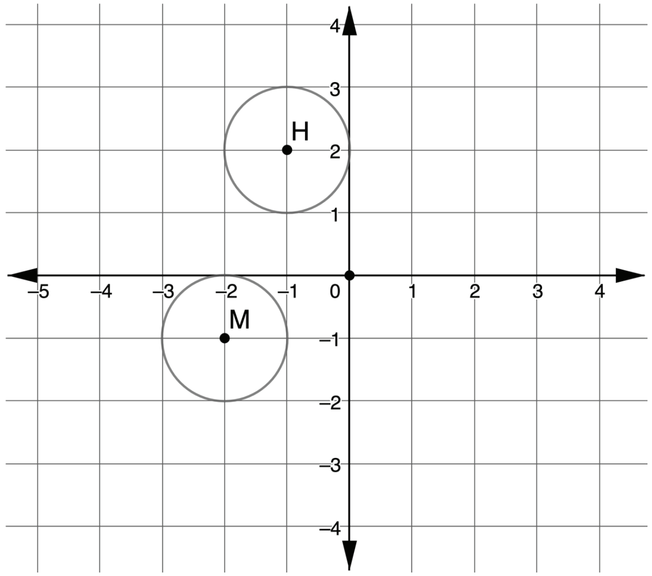
1. What do you notice?

2. What do you wonder?

6.H1.2 Exploration: Are Circles Similar?

Answer the following questions with your partner.

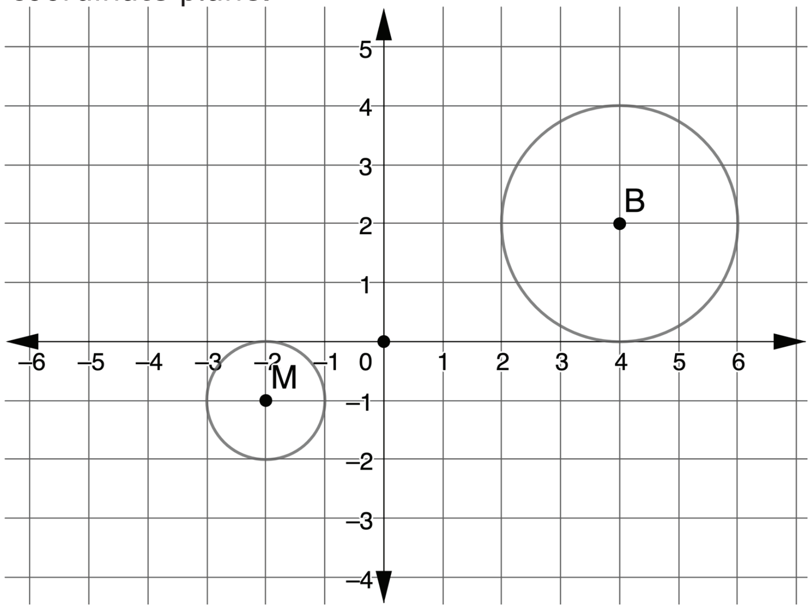
1. Circle M and Circle H are shown on the coordinate plane.



   a. How can Circle M be mapped onto Circle H?

   b. What can you conclude about Circle M and Circle H?

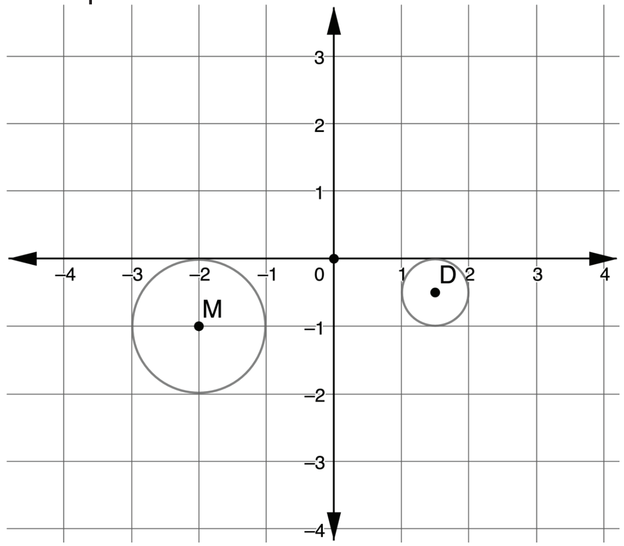
2. Circle M and Circle B are shown on the coordinate plane.



a. How can Circle M be mapped onto Circle B?

b. What can you conclude about Circle M and Circle B?

3. Circle M and Circle D are shown on the coordinate plane.



a. How can Circle M be mapped onto Circle D?

b. What can you conclude about Circle M and Circle D?

4. What can you conclude about Circle M, Circle H, Circle B, and Circle D?

5. Complete the statement.

○ All
○ Some
○ None

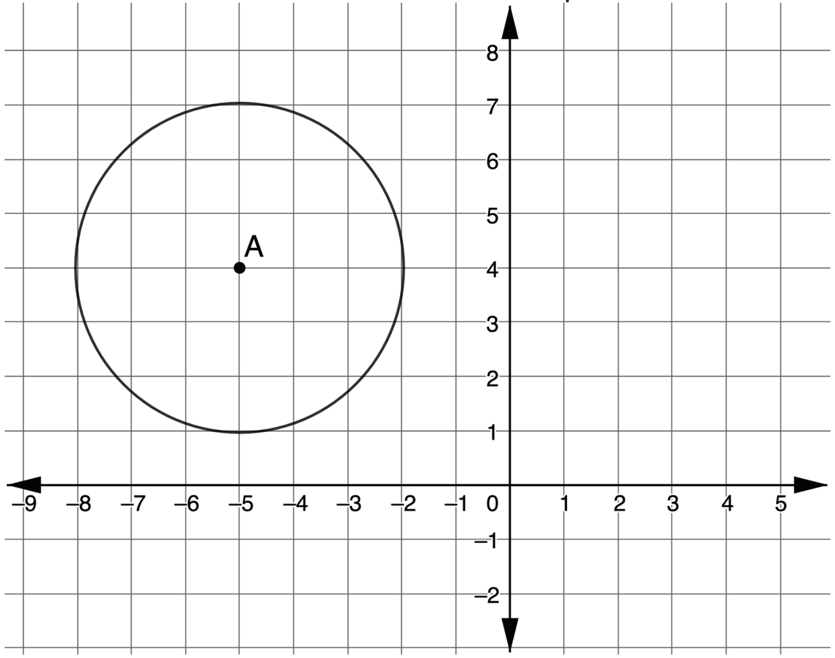
circles are

○ congruent
○ similar

because any circle can be mapped to another circle using similarity transformations.

6.H1.3 Your Turn!

1. Circle A is shown on the coordinate plane.



   a. Draw Circle P, the result of a translation 5 units right and 2 units down and a dilation of $\frac{1}{2}$ about the center of Circle P.

   b. Explain how Circle A and Circle P are similar. Include the word "transformation" in your explanation.

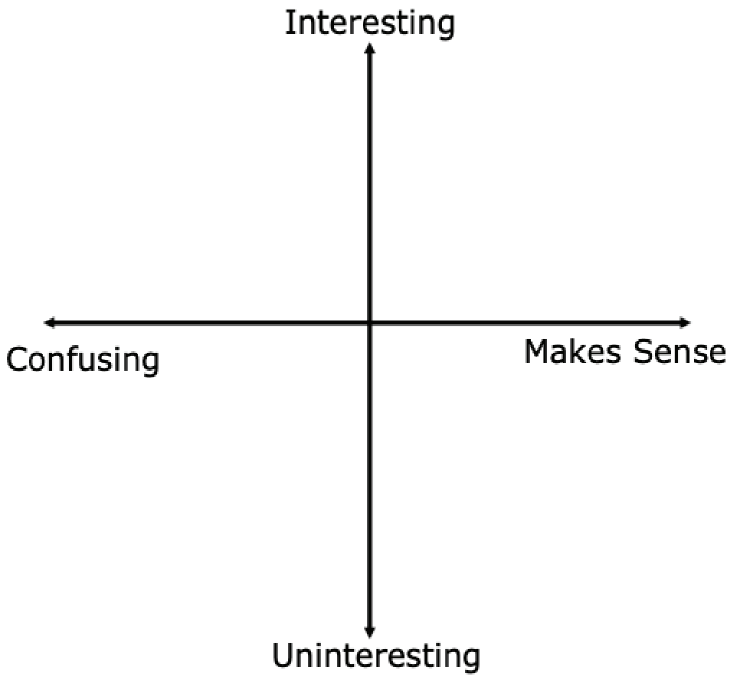
2. Lana and Henry are writing notes from today's lesson.

Lana's Notes	Henry's Notes
All circles are similar; therefore, all circles are also congruent.	All circles are similar, but not all circles are congruent.

   Whose notes do you agree with? Why?

6.H1.4 Wrap-Up: Reflection

1. Plot a point on the gird below to indicate your level of interest and understanding for today's learning target.



Explain the location of your point.

Unit 6, Lesson H2: Proofs by Contradiction

Benchmark

MA.912.LT.4.8 Construct proofs, including proofs by contradiction.

Learning Target

☐ I can use proof by contradiction to justify that all circles are similar.